

ABERDEEN, SD, USA

WMO: 726590

Lat: 45.443N

Lon: 98.413W

Elev: 1297

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14929

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-12.0	-22.8	1.6	-16.2	-17.8	2.2	-11.1	33.1	14.8	29.0	14.6	6.6	320	0.670

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	91.0	73.1	87.9	71.6	85.2	69.9	76.8	86.4	74.5	84.2	72.6	82.0	13.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	132.3	83.2	71.4	121.6	80.5	69.2	112.6	78.7	41.2	85.8	39.0	84.2	37.1	82.0	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
28.4	25.3	22.7	DB	-27.9	97.2	7.6	4.7	-33.4	100.5	-37.9	103.3	-42.2	105.9	-47.7	109.3	
			WB	-26.5	80.2	6.4	2.1	-31.1	81.7	-34.9	82.9	-38.5	84.1	-43.1	85.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	12.2	16.5	29.6	44.1	56.7	67.2	72.1	69.2	60.8	46.3	30.8	17.8
	DBStd	23.34	13.80	13.89	13.30	10.20	8.83	6.92	5.85	5.88	8.72	9.47	11.43	12.35
	HDD50	4785	1171	937	640	227	37	0	0	0	12	184	580	997
	HDD65	8349	1636	1357	1099	629	283	54	11	26	182	584	1026	1462
	CDD50	2504	0	0	7	49	244	516	685	595	336	68	4	0
	CDD65	594	0	0	0	1	26	120	231	156	56	4	0	0
	CDH74	6419	0	0	4	53	402	1327	2378	1548	642	65	0	0
	CDH80	2195	0	0	0	10	122	450	901	501	198	13	0	0
Wind	WSAvg	10.4	10.4	10.7	11.3	12.2	11.7	10.3	8.7	8.5	9.8	10.6	10.4	10.3
Precipitation	PrecAvg	21.6	0.5	0.6	1.3	2.1	3.0	3.8	3.1	2.4	2.0	1.6	0.7	0.4
	PrecMax	30.2	1.4	2.0	4.4	7.9	7.4	9.3	9.8	5.9	5.5	7.3	3.0	1.4
	PrecMin	10.0	0.0	0.0	0.0	0.1	0.3	0.5	0.4	0.2	0.0	0.0	0.0	0.0
	PrecStd	4.4	0.4	0.4	1.0	1.6	1.7	2.2	1.9	1.4	1.5	1.5	0.7	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	52.3	70.0	80.4	88.5	93.8	96.1	92.5	91.0	81.1	66.7	49.9
		MCWB	38.5	44.1	53.9	58.6	66.2	71.8	75.5	74.7	70.8	62.5	52.4	41.5
	2%	DB	38.7	44.5	60.9	73.1	83.3	88.7	91.3	88.9	85.5	73.9	58.4	42.8
		MCWB	34.8	39.1	49.9	54.6	63.8	70.7	74.9	73.1	68.8	58.9	48.0	36.3
	5%	DB	34.9	38.9	54.2	67.5	78.4	85.1	88.1	85.3	81.2	67.8	52.0	37.7
		MCWB	32.3	35.2	46.1	51.9	61.7	68.5	73.0	71.1	65.7	55.2	43.9	33.9
	10%	DB	31.5	35.3	47.8	62.3	73.5	81.6	85.1	82.4	77.0	62.5	46.8	34.0
		MCWB	29.4	32.7	41.6	49.4	59.1	66.7	71.6	69.3	64.0	52.6	40.4	31.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.1	45.7	57.4	61.0	69.9	76.7	80.5	78.5	73.8	65.5	54.2	41.6
		MCDB	44.5	51.5	64.8	76.2	83.5	87.9	89.7	87.9	85.2	75.8	63.9	49.2
	2%	WB	34.9	39.1	51.0	56.2	66.6	73.8	77.8	75.3	71.1	60.6	48.9	36.9
		MCDB	38.2	43.8	59.1	69.8	79.3	84.4	87.1	84.7	82.2	70.3	57.9	41.5
	5%	WB	32.5	35.5	46.2	53.3	63.6	71.2	75.6	73.3	68.7	56.8	44.4	34.0
		MCDB	35.0	38.7	54.6	64.5	75.0	81.6	85.1	82.5	78.1	65.9	51.9	37.5
	10%	WB	29.6	32.8	41.5	50.5	60.7	68.8	73.4	71.2	66.1	53.2	40.5	31.6
		MCDB	31.5	35.4	47.8	61.0	71.0	78.7	82.6	79.6	74.4	61.4	46.0	34.2
Mean Daily Temperature Range	5% DB	MDBR	18.9	19.7	19.6	23.1	23.2	22.3	22.6	23.3	25.0	22.9	20.8	18.0
		MCDBR	22.0	22.2	30.2	33.6	31.1	27.0	26.1	26.6	29.7	32.2	29.6	23.8
		MCWBR	18.7	17.0	19.4	18.0	15.5	12.3	12.7	13.2	14.7	18.5	19.5	18.2
	5% WB	MCDBR	21.5	21.0	29.2	29.5	26.6	22.8	22.8	23.8	24.2	28.6	28.9	22.5
		MCWBR	18.3	16.4	19.4	16.6	14.4	12.2	12.7	13.0	13.6	17.5	19.5	17.8
Clear-Sky Solar Irradiance	taub	0.266	0.282	0.304	0.353	0.378	0.383	0.387	0.395	0.358	0.318	0.285	0.265	
	taud	2.417	2.406	2.450	2.328	2.339	2.372	2.368	2.326	2.408	2.487	2.476	2.437	
	Ebn at Noon	272	288	296	287	282	280	278	271	272	271	263	260	
	Edh at Noon	22	28	30	38	39	38	37	37	32	25	21	20	
All-Sky Solar Radiation	RadAvg	563	903	1236	1578	1828	2055	2121	1770	1379	889	587	434	
	RadStd	50	94	107	116	148	130	85	150	86	109	65	28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page